

Viswanathan T J

Software Engineer

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SUMMARY

Backend-leaning full-stack engineer with **3+ years at Zoho**, building Python and Java services — **REST APIs, microservices**, and a shared API gateway — with React/Next.js frontends. Ships and operates production systems end to end with **Docker, CI/CD**, and self-hosted infrastructure.

SKILLS

PROGRAMMING

PythonJavaJavaScriptShell Scripting

BACKEND

FastAPIFlaskDjangoSpring BootREST APIMicroservices

FRONTEND

ReactNext.jsHTMLCSS

DATA & AUTOMATION

NumPyPandasAutomation Scripts

CLOUD & DEVOPS

DockerKubernetesGoogle CloudGitHub ActionsLinux

DATABASES

PostgreSQLMySQLMongoDBRedis

EXPERIENCE

Software Development Engineer 06/2023 – Present
Zoho Corporation · Chennai

- Built a CLI-based **server configuration and deployment automation** tool in Python and Shell, simplifying setup and releases for the whole team.
- Cut developer cycle time by **~40%** by streamlining environment setup, code changes, and testing loops.
- Developed backend components for a shared **API gateway** integrating **10+ external systems**, standardizing CRUD operations and reducing maintenance complexity.
- Implemented **real-time data synchronization** across interconnected services, improving data accuracy and eliminating manual intervention.

Software Development Engineer — Intern 02/2021 – 06/2023
Zoho Corporation · Chennai

- Co-built a full-stack **project management tool** adopted by **200+ internal users**, developing backend APIs and UI components.
- Diagnosed and fixed **30+ production issues**, strengthening debugging and root-cause analysis skills.
- Automated repetitive integration steps, significantly reducing service **integration time**.

PROJECTS

Academic Suite — Exam Controller 2026

FastAPI · React/TypeScript · PostgreSQL · Redis · Docker

- Built a full-stack platform automating **marksheets, hall-ticket, and report generation** for PG/UG/Diploma programs — pixel-accurate PDFs (A4/Legal/A5) rendered from Jinja2 templates.
- Designed a unified **grading engine** covering three academic systems with **Strategy + Factory** patterns, keeping program rules isolated and extensible.
- Served two role-aware React frontends from one **FastAPI** backend: staff portal with granular **RBAC** and time-windowed marks entry; student portal with tab-scoped **JWT** and AES-GCM-encrypted sessions.
- Modeled the data layer on **PostgreSQL** with typed SQLAlchemy 2.0 and a repository pattern; **Dockerized** and self-deployed on a VPS with backups and monitoring.

Marksheet Management System 07/2021 – 10/2021

- Flask** + Bootstrap web app generating student marksheets — in **production use for 5 years** at the institution before being rebuilt as Academic Suite.

Password Manager — Web & Desktop 09/2020 – 12/2020

- Python password manager with **PyQt5** desktop and **Flask** web clients, backed by MongoDB/Firebase.

EDUCATION

MCA — Master of Computer Applications
Kongu Engineering College, Erode · 10/2021 – 04/2023

B.Sc Computer Science
Thiagarajar College, Madurai · 04/2018 – 05/2021

CERTIFICATES

RHCSA — Red Hat Certified System Administrator (EX200)
ID: 210-048-773 · 2021

AI Agents Intensive Course
Google · 2024

LANGUAGES

Tamil English